The Causal Higgs Mechanism at the Schwarzschild Horizon:

Mass Generation, Dissolution of Matter, and the Destruction of Causality

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Abstract

We examine the causal structure of the Schwarzschild interior and establish the following rigorous result: on any constant-r hypersurface inside the event horizon, the induced metric is positive-definite (Riemannian), and no two distinct events on such a surface can be connected by a causal curve. This elementary observation has a profound consequence: spatially extended objects, whose internal coherence requires causal contact between their constituent parts, cannot maintain their identity in the interior. We identify *nothingness* with this state of causal disconnectedness on spatial slices. We then construct a local $U(1)_{\mathcal{C}}$ gauge theory in which a causal scalar field, coupled to a gauge field that mediates causal interactions, undergoes spontaneous symmetry breaking at the horizon. The resulting Higgs mechanism generates mass at the boundary at a rate proportional to M^{-2}, where M is the black hole mass. We derive a causality destruction theorem: the integrated causal content of the exterior decreases monotonically with increasing horizon area. The Bekenstein–Hawking entropy is reinterpreted as a count of causal degrees of freedom that have been converted to nothingness. All results pertain to the isolated Schwarzschild black hole.

Keywords: Schwarzschild spacetime, causal structure, Higgs mechanism, event horizon, information loss, nothingness

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1 Introduction

The Schwarzschild solution describes the vacuum spacetime exterior and interior to a spherically symmetric, non-rotating, uncharged mass. At the gravitational radius $r_s = 2GM/c^2$, the standard Schwarzschild coordinates exhibit a coordinate singularity where the roles of the temporal and radial coordinates exchange [1]. While this singularity is removable by Kruskal–Szekeres or other regular coordinates [2], the physical content of the exchange—that the interior possesses a fundamentally different causal structure from the exterior—is coordinate-invariant.

In this paper we take this difference seriously. We begin with a precise analysis of the causal structure on spatial hypersurfaces inside and outside the horizon (section 2). The central observation is elementary but has consequences that appear to be underappreciated: inside the horizon, the constant-r surfaces (which are the natural "equal-time" surfaces, since r is the timelike coordinate) are Riemannian. Every pair of distinct events on such a surface is spacelike separated. No causal curve confined to or departing from and returning to such a surface exists. We call this property *spatial causal disconnectedness*.

This leads directly to a physical consequence (section 3): any spatially extended object—an electron, a proton, an atom, a field configuration—requires causal contact between its spatially separated parts to maintain its identity and coherence. Inside the horizon, this contact is impossible on any equal-time surface. Extended objects cannot exist as coherent entities in the Schwarzschild interior.

We then propose (sections 5 and 6) that this transition in causal structure is governed by a spontaneous symmetry breaking of a local $U(1)_{\mathcal{C}}$ gauge symmetry. A causal scalar field acquires a position-dependent vacuum expectation value that vanishes at the horizon, triggering a Higgs mechanism. The mass generated by this mechanism, and the rate of its generation, are computed in section 7. The consequences for information theory and the area theorem are derived in sections 8 and 9.

We make the following identification throughout:

> *Nothingness is causal disconnectedness on spatial hypersurfaces—the condition under which no spatially extended physical object can maintain its coherence.*

2 Causal Structure of the Schwarzschild Interior

2.1 The Schwarzschild metric

The Schwarzschild line element in coordinates (t, r, θ, φ) is

$$ds^2 = -f(r)\, c^2\, dt^2 + f(r)^{-1}\, dr^2 + r^2\, d\Omega^2\,, \tag{1}$$

where $f(r) = 1 - r_s/r$ and $d\Omega^2 = d\theta^2 + \sin^2\theta\, d\varphi^2$.

For $r > r_s$: $f > 0$, so $g_{tt} < 0$ (timelike) and $g_{rr} > 0$ (spacelike). The coordinate t labels time and r labels a spatial direction.

For $r < r_s$: $f < 0$, so $g_{tt} > 0$ (spacelike) and $g_{rr} < 0$ (timelike). The coordinate r now labels time and t labels a spatial direction.

At $r = r_s$: $f = 0$. The surface is null.

2.2 Spatial hypersurfaces and their signatures

In the exterior ($r > r_{\rm s}$), the natural spatial hypersurfaces are $\Sigma_t = \{t = \text{const}\}$. The induced metric on Σ_t is

$$ds^2\big|_{\Sigma_t} = f(r)^{-1}\, dr^2 + r^2\, d\Omega^2\,, \tag{2}$$

which is positive-definite for $r > r_{\rm s}$. This is a Riemannian metric, as expected for a spatial slice.

In the interior ($r < r_{\rm s}$), the timelike coordinate is r. The natural equal-time hypersurfaces are $\Sigma_r = \{r = \text{const}\}$. The induced metric on Σ_r is

$$ds^2\big|_{\Sigma_r} = |f(r)|\, c^2\, dt^2 + r^2\, d\Omega^2\,, \tag{3}$$

where we have used $-f(r) = |f(r)|$ for $r < r_{\rm s}$. Since $|f(r)| > 0$ and $r^2 > 0$, this induced metric is *also positive-definite*. It is Riemannian.

Proposition 2.1 (Spatial causal disconnectedness of the interior)**.** *Let $r_0 < r_{\rm s}$ and let p, q be two distinct events on Σ_{r_0}. Then p and q are spacelike separated: there exists no causal (timelike or null) curve connecting them that remains on Σ_{r_0}.*

Proof. The tangent vector to any curve on Σ_{r_0} has $dr = 0$ and is therefore a linear combination of ∂_t, ∂_θ, ∂_φ. Its norm is $|f|\, c^2\, \dot{t}^2 + r_0^2\, \dot{\theta}^2 + r_0^2 \sin^2\theta\, \dot{\varphi}^2 > 0$ for any non-trivial tangent vector. Every curve on Σ_{r_0} is spacelike. □

2.3 Comparison: exterior vs. interior

In both regions, the equal-time spatial hypersurface is Riemannian, and events at the same time but different spatial locations are spacelike separated. However, the crucial difference lies in the *ability of spatial points to communicate through time evolution.*

Exterior. Consider two events p and q on Σ_t at different spatial locations. Although they are spacelike separated on Σ_t, a signal sent from p at time t can reach the spatial location of q at some later time $t + \Delta t$. The signal departs Σ_t, propagates through the (+)-time direction (increasing t), and arrives at q's location. The reverse is also possible. Both share unlimited common causal future ($t \to \infty$) and past ($t \to -\infty$).

Interior. Consider two events p and q on Σ_{r_0} at different locations $(t_p, \theta_p, \varphi_p)$ and $(t_q, \theta_q, \varphi_q)$. A signal from p propagates in the timelike direction, which is the direction of decreasing r (toward the singularity). This signal can reach the angular/t-coordinates of q, but only at some $r_1 < r_0$—a *later* time. It can never return to r_0. Event q, independently, also evolves to smaller r. The two signals can *meet* at some $r_1 < r_0$, but:

(i) The meeting occurs at a later time (smaller r), not at the original time r_0.

(ii) The common causal future is bounded: it terminates at $r = 0$ (the singularity).

(iii) No return signal from the meeting point can reach either p or q at their original time r_0.

Definition 2.2 (Causal coherence time)**.** *The **causal coherence time** at radius r inside the horizon is the proper time remaining until the singularity:*

$$\tau_{\rm coh}(r) = \frac{r_{\rm s}}{c}\left[\arccos\!\left(\sqrt{\frac{r}{r_{\rm s}}}\right) + \sqrt{\frac{r}{r_{\rm s}}\left(1 - \frac{r}{r_{\rm s}}\right)}\,\right], \tag{4}$$

for a radially free-falling observer from rest at $r_{\rm s}$.

At $r = r_s$: $\tau_{\rm coh} = \pi r_s/(2c)$. At $r = 0$: $\tau_{\rm coh} = 0$.

The maximum spatial distance over which causal contact can be established at radius r is bounded by $c\,\tau_{\rm coh}(r)$. As $r \to 0$, this coherence length vanishes, and even infinitesimally separated points lose all causal contact.

3 Dissolution of Extended Objects

3.1 The coherence requirement

We now state a physical principle that is implicit in all of field theory but is rarely made explicit.

Postulate 1 (Causal coherence requirement)**.** *A spatially extended physical object of characteristic size L can exist as a coherent entity only if all pairs of points within its spatial extent can exchange causal signals within a time of order L/c—that is, only if the object fits within a causal diamond of size $\geq L$.*

This is not a new physical assumption. It is the content of cluster decomposition in quantum field theory: for a composite system to behave as a single entity rather than as independent subsystems, its parts must be in causal contact. An electron with Compton wavelength $\lambda_C = \hbar/(m_e c)$ maintains its identity because all points within λ_C can exchange virtual photons and maintain the coherent quantum state.

3.2 Failure of coherence in the interior

Inside the horizon, two points on the same equal-time surface Σ_r are spacelike separated (theorem 2.1). They cannot exchange a causal signal at that instant. A signal from one reaches the other's location only at a later time (smaller r), but the other has also moved to smaller r in the interim. The coherence of the original configuration is never restored.

More precisely, let an object of proper size L exist on the surface Σ_{r_0} with $r_0 < r_s$. Two constituent parts separated by proper distance L on Σ_{r_0} can only meet causally at some $r_1 < r_0$. At r_1, the object's size has changed (due to the evolving geometry), the meeting occurs at a different proper time for each part, and the coherence established at r_0 has been disrupted. As r decreases, the coherence time $\tau_{\rm coh}(r)$ shrinks, and eventually:

$$c\,\tau_{\rm coh}(r) < L \quad \Longrightarrow \quad \text{coherence is impossible.} \tag{5}$$

For an electron, $L = \lambda_C \approx 3.86 \times 10^{-13}$ m. The dissolution occurs when

$$\tau_{\rm coh}(r_{\rm diss}) = \frac{\lambda_C}{c} = \frac{\hbar}{m_e c^2} \approx 1.29 \times 10^{-21}\ \text{s}\,. \tag{6}$$

For a stellar-mass black hole ($r_s \sim 3$ km), $\pi r_s/(2c) \sim 10^{-5}$ s, so the electron survives for most of the interior. But the point of principle stands: at sufficiently small r, dissolution is inevitable for *every* extended object. At $r = 0$, $\tau_{\rm coh} = 0$, and no object of any size survives.

3.3 The hierarchy of dissolution

Different objects dissolve at different radii, set by their coherence scale L:

$$c\,\tau_{\mathrm{coh}}(r_{\mathrm{diss}}) = L\,. \tag{7}$$

Large objects (stars, planets) dissolve almost immediately inside the horizon. Atoms dissolve when the coherence length drops below the Bohr radius. Nuclei dissolve at the nuclear scale. Elementary particles dissolve at their Compton wavelength. At the singularity, the coherence length is zero, and nothing survives.

This is fundamentally different from tidal destruction (spaghettification). Tidal forces arise from the Riemann curvature and scale as M/r^3; for a supermassive black hole, tidal forces at the horizon are negligible. The dissolution described here is *causal*, not tidal: it occurs because the background spacetime cannot support the causal connections that define the object's identity, regardless of the magnitude of tidal forces.

Remark 3.1. *The horizon of a supermassive black hole has negligible tidal forces, and a freely falling observer notices nothing special at the crossing [11]. Yet the causal dissolution has already begun: from that moment, the observer's spatial extent is on a Riemannian hypersurface where different parts cannot communicate at equal time. The dissolution is gentle for large black holes but inexorable.*

4 Nothingness as Causal Disconnectedness

We are now in a position to give nothingness a precise definition.

Definition 4.1 (Nothingness)**.** *A spacetime region $\mathcal{U}$ is in a state of* ***nothingness*** *if no spatially extended physical object can maintain causal coherence within $\mathcal{U}$. Formally: for every $L > 0$, there exists no spacelike hypersurface $\Sigma \subset \mathcal{U}$ on which a causal diamond of spatial diameter L can be constructed within the remaining causal future.*

By this definition:

- The singularity $r = 0$ is absolute nothingness: $\tau_{\mathrm{coh}} = 0$, no coherence at any scale.
- The Schwarzschild interior at any $r < r_{\mathrm{s}}$ is partial nothingness: objects with $L > c\,\tau_{\mathrm{coh}}(r)$ cannot exist.
- The Schwarzschild exterior is not nothingness: $\tau_{\mathrm{coh}} = \infty$, all objects can maintain coherence indefinitely.
- The horizon $r = r_{\mathrm{s}}$ is the boundary: the onset of nothingness.

Nothingness is not the absence of the spacetime manifold. The metric, the curvature, the topology all persist. Nothingness is the absence of the causal infrastructure required for physical objects to exist. It is spacetime emptied of the capacity for being.

5 The Causal Gauge Theory

5.1 Motivation

The analysis of sections 2 and 3 is purely kinematic: it follows from the Schwarzschild metric alone. We now seek a *dynamical* description of the transition from causal coherence (exterior) to causal disconnectedness (interior). We propose that this transition is governed by a local gauge theory.

The physical motivation is the following. In the exterior, causal influence propagates between spacetime points via null and timelike curves. This propagation can be modeled as a gauge-mediated interaction: a "causal mediator" field $\mathcal{A}_\mu$ carries causal influence from event to event, much as the photon carries electromagnetic influence. If this mediator is massless, causal influence is long-range (limited only by the light cone). If the mediator acquires mass through a Higgs mechanism, causal influence becomes short-range, decaying as $e^{-m_{\mathcal{A}} r}$. At the horizon, if $m_{\mathcal{A}}$ diverges (or the condensate vanishes entirely), causal propagation ceases.

5.2 Field content

We introduce:

(i) A complex scalar field φ (the **causal scalar**), charged under a local $U(1)_{\mathcal{C}}$ symmetry.

(ii) A gauge field $\mathcal{A}_\mu$ (the **causal mediator**).

The gauge transformation acts as

$$\varphi \to e^{i\alpha(x)}\varphi\,, \qquad \mathcal{A}_\mu \to \mathcal{A}_\mu + \frac{1}{g_{\mathcal{C}}}\partial_\mu \alpha\,, \tag{8}$$

with gauge coupling $g_{\mathcal{C}}$. The covariant derivative and field strength are

$$D_\mu \varphi = \partial_\mu \varphi - i g_{\mathcal{C}} \mathcal{A}_\mu \varphi\,, \qquad \mathcal{F}_{\mu\nu} = \partial_\mu \mathcal{A}_\nu - \partial_\nu \mathcal{A}_\mu\,. \tag{9}$$

5.3 Lagrangian

The causal sector Lagrangian density is

$$L_{\mathcal{C}} = -\frac{1}{4}\mathcal{F}_{\mu\nu}\mathcal{F}^{\mu\nu} + (D_\mu\varphi)^*(D^\mu\varphi) - V(|\varphi|)\,, \tag{10}$$

with the standard symmetry-breaking potential

$$V(|\varphi|) = -\mu^2|\varphi|^2 + \lambda|\varphi|^4\,, \qquad \mu^2 > 0,\ \lambda > 0\,. \tag{11}$$

The vacuum manifold is the circle $|\varphi| = v \equiv \mu/\sqrt{2\lambda}$.

5.4 Coupling to the Schwarzschild geometry

The key physical input is the coupling between the causal scalar and the background geometry. We *postulate* that the effective vacuum expectation value of the causal scalar is determined by the metric signature:

Postulate 2 (Geometric coupling)**.** *The vacuum expectation value of the causal scalar in the Schwarzschild geometry is*

$$\langle|\varphi(r)|\rangle = v\,\Theta(r - r_{\rm s})\sqrt{f(r)}\,, \tag{12}$$

where Θ *is the Heaviside step function and* $f(r) = 1 - r_{\rm s}/r$*.*

This is the simplest ansatz consistent with the following requirements:

(a) $\langle|\varphi|\rangle \to v$ as $r \to \infty$: full condensate far from the black hole.

(b) $\langle|\varphi|\rangle \to 0$ as $r \to r_{\rm s}^{+}$: the condensate vanishes at the horizon.

(c) $\langle|\varphi|\rangle = 0$ for $r \le r_{\rm s}$: no condensate in the interior.

Requirement (c) is the content of our causal analysis: the interior is nothingness, and the causal condensate is identically zero there. The square root of $f(r)$ is chosen because it is the simplest function of the Schwarzschild geometry that vanishes at $r_{\rm s}$ and approaches unity at infinity; it equals $\sqrt{-g_{tt}/c^2}$, the redshift factor.

6 The Causal Higgs Mechanism at the Horizon

6.1 Gauge boson mass

Expanding φ around the local VEV in unitary gauge (Goldstone boson absorbed by $\mathcal{A}_\mu$), the covariant derivative term in (10) generates a position-dependent gauge boson mass:

$$m_{\mathcal{A}}^2(r) = 2\,g_{\mathcal{C}}^2\,\langle|\varphi(r)|\rangle^2 = 2\,g_{\mathcal{C}}^2\,v^2\,f(r)\,\Theta(r - r_{\rm s})\,. \tag{13}$$

Exterior ($r > r_{\rm s}$): $m_{\mathcal{A}}(r) = g_{\mathcal{C}} v\sqrt{2f(r)} > 0$. The causal mediator is massive, and causal interactions have a finite range $\sim \hbar/(m_{\mathcal{A}}c)$. Far from the black hole, $m_{\mathcal{A}} \to g_{\mathcal{C}} v\sqrt{2} \equiv m_\infty$, and causality has the standard long-range character (in the limit $g_{\mathcal{C}} \to 0$, or $m_\infty \to 0$, the range becomes infinite, recovering the usual picture).

At the horizon ($r = r_{\rm s}$): $m_{\mathcal{A}} = 0$. The condensate vanishes. The $U(1)_{\mathcal{C}}$ symmetry is *restored*, not broken. In the Higgs mechanism, the restoration of the symmetry corresponds to the loss of the condensate and the disappearance of the mechanism that gave the gauge boson mass. The causal mediator becomes massless *in the trivial sense*: there is no condensate for it to interact with. Causal mediation ceases.

Interior ($r < r_{\rm s}$): $\langle|\varphi|\rangle = 0$ identically. There is no condensate. The causal mediator has no mass, no condensate to couple to, and no role. The interior is a "causal vacuum desert"—a region devoid of the infrastructure for causal interaction.

6.2 The Meissner analogy

The closest known analogue is the Meissner effect in superconductivity [6]:

Superconductor	**Schwarzschild spacetime**
Cooper pair condensate $\langle\psi\rangle \neq 0$	Causal condensate $\langle\lvert\varphi\rvert\rangle \neq 0$
Photon mass $m_\gamma = g_e\lvert\langle\psi\rangle\rvert$	Mediator mass $m_{\mathcal{A}} = g_{\mathcal{C}}\sqrt{2}\,\lvert\langle\varphi\rangle\rvert$
Magnetic field expelled	—
Normal phase: $\langle\psi\rangle = 0$, $m_\gamma = 0$	Interior: $\langle\lvert\varphi\rvert\rangle = 0$, $m_{\mathcal{A}} = 0$

In a superconductor, the normal phase ($\langle\psi\rangle = 0$) has no condensate and a massless photon—electromagnetic fields penetrate freely. The superconducting phase ($\langle\psi\rangle \neq 0$) has a massive photon—magnetic fields are expelled.

At the black hole horizon, the analogy is inverted in physical interpretation but structurally identical. The exterior ($\langle\lvert\varphi\rvert\rangle \neq 0$) supports causal propagation through the massive mediator. The interior ($\langle\lvert\varphi\rvert\rangle = 0$) has no condensate and no causal propagation—not because signals are "expelled," but because the mechanism enabling their propagation (the condensate) has evaporated.

6.3 The Higgs (radial) mode

The fluctuation of $|\varphi|$ around the local VEV gives a scalar mode with position-dependent mass:

$$m_h^2(r) = 4\lambda\,\langle|\varphi(r)|\rangle^2 = 4\lambda\,v^2\,f(r)\,\Theta(r-r_{\rm s}) = 2\mu^2\,f(r)\,\Theta(r-r_{\rm s})\,. \tag{14}$$

At the horizon, $m_h \to 0$: both the gauge boson and the Higgs mode become massless at the phase boundary. This simultaneous vanishing is the hallmark of the critical point of the causal phase transition.

7 Mass Generation at the Horizon Boundary

7.1 Energy of the causal condensate gradient

The VEV profile (12) is not constant; it varies with r, and this variation stores energy. The gradient energy density in the causal sector is

$$\rho_{\rm grad}(r) = \frac{1}{2}\left(\frac{d\langle|\varphi|\rangle}{dr}\right)^2 = \frac{v^2 r_{\rm s}^2}{8\,r^4\,f(r)}\,, \qquad r > r_{\rm s}\,, \tag{15}$$

where we have used

$$\frac{d\langle|\varphi|\rangle}{dr} = \frac{v\,r_{\rm s}}{2\,r^2\,\sqrt{f(r)}}\,. \tag{16}$$

This density diverges as $r \to r_{\rm s}^+$ due to the factor $f(r)^{-1}$.

The total energy stored in the condensate gradient, integrated with the proper volume element, is

$$E_{\rm grad} = \int_{r_{\rm s}}^{\infty} \rho_{\rm grad}(r)\,\frac{4\pi r^2}{\sqrt{f(r)}}\,dr = \frac{\pi v^2 r_{\rm s}^2}{2}\int_{r_{\rm s}}^{\infty}\frac{dr}{r^2\,f(r)^{3/2}}\,. \tag{17}$$

Substituting $u = r/r_\mathrm{s} - 1$:

$$E_\mathrm{grad} = \frac{\pi v^2}{2r_\mathrm{s}} \int_0^\infty \frac{(1+u)^{-2}}{(u/(1+u))^{3/2}} \frac{du}{1} = \frac{\pi v^2}{2r_\mathrm{s}} \int_0^\infty \frac{(1+u)^{-1/2}}{u^{3/2}}\, du\,. \tag{18}$$

This integral diverges at $u = 0$, reflecting the concentration of energy at the horizon. With a physical cutoff at the Planck length ℓ_P (i.e., $u_\mathrm{min} = \ell_\mathrm{P}/r_\mathrm{s}$), the dominant contribution is

$$E_\mathrm{grad} \sim \frac{\pi v^2}{r_\mathrm{s}} \sqrt{\frac{r_\mathrm{s}}{\ell_\mathrm{P}}} = \pi v^2 \sqrt{\frac{1}{\ell_\mathrm{P} r_\mathrm{s}}}\, r_\mathrm{s}\,. \tag{19}$$

7.2 Surface energy density

The energy is concentrated in a thin shell of thickness $\sim \ell_\mathrm{P}$ near the horizon. The surface energy density is

$$\sigma = \frac{E_\mathrm{grad}}{4\pi r_\mathrm{s}^2} \sim \frac{v^2}{4r_\mathrm{s}} \sqrt{\frac{r_\mathrm{s}}{\ell_\mathrm{P}}}\,. \tag{20}$$

7.3 Self-consistency: fixing the causal VEV

For the causal Higgs mechanism to be the source of the black hole's gravitational mass, we require $E_\mathrm{grad} = Mc^2$:

$$\pi v^2 \sqrt{\frac{1}{\ell_\mathrm{P} r_\mathrm{s}}}\, r_\mathrm{s} = \frac{r_\mathrm{s} c^4}{2G}\,, \tag{21}$$

using $M = r_\mathrm{s} c^2/(2G)$. This gives

$$v^2 = \frac{c^4}{2\pi G} \sqrt{\ell_\mathrm{P} r_\mathrm{s}}\,. \tag{22}$$

This depends on r_s (and hence M), indicating that the causal VEV is not a universal constant but is fixed by the self-consistent coupling between the causal sector and gravity. For a Planck-mass black hole ($r_\mathrm{s} \sim \ell_\mathrm{P}$), $v^2 \sim c^4 \ell_\mathrm{P}/(\pi G) = c^4/(\pi G/\ell_\mathrm{P}) \sim c\hbar/(\pi G) \cdot c^3 = M_\mathrm{P}^2 c^2/\pi$, and the VEV is of order the Planck scale, as expected.

7.4 The mass creation rate

The dynamical rate of mass generation at the horizon is governed by the surface gravity κ, which controls the rate of change of the causal field at the boundary:

$$\kappa = \frac{c^4}{4GM} = \frac{c^2}{2r_\mathrm{s}}\,. \tag{23}$$

The surface gravity has dimensions of acceleration and is the only intrinsic geometric quantity characterizing the Schwarzschild horizon. On dimensional grounds, the mass creation rate takes the form

$$\dot{M}_\mathrm{create} = \beta\, \kappa^2\,, \tag{24}$$

where β is a coupling constant with dimensions [mass^3 length^{-2} time] that parameterizes the efficiency of the Higgs mechanism at converting causal vacuum energy into gravitational mass. Substituting:

$$\boxed{\dot{M}_\mathrm{create} = \frac{\beta\, c^8}{16\, G^2\, M^2}\,, \qquad \dot{M}_\mathrm{create} \propto M^{-2}\,.} \tag{25}$$

Smaller black holes generate mass at their horizons at a dramatically higher rate. This is a direct consequence of the surface gravity being larger for smaller black holes.

7.5 Horizon growth

Since $r_\mathrm{s} = 2GM/c^2$:

$$\dot{r}_\mathrm{s} = \frac{2G}{c^2}\dot{M}_\mathrm{create} = \frac{\beta\, c^6}{8G\, r_\mathrm{s}^2}\,, \qquad \dot{r}_\mathrm{s} \propto r_\mathrm{s}^{-2}\,. \tag{26}$$

Integrating:

$$r_\mathrm{s}^3(t) = r_\mathrm{s}^3(0) + \frac{3\beta\, c^6}{8G}\, t\,. \tag{27}$$

At late times, $r_\mathrm{s} \propto t^{1/3}$. The growth is self-regulating: as the black hole grows, the creation rate decreases.

8 Information Nullification

8.1 Causal information content

We define information in terms of the causal condensate, which determines the capacity of a region to support coherent physical objects and causal relationships.

Definition 8.1 (Causal information content)**.** *The causal information content of a spacelike hypersurface Σ in the exterior is*

$$\mathcal{I}[\Sigma] = \frac{1}{4\ell_\mathrm{P}^2}\int_\Sigma \frac{\langle|\varphi(r)|\rangle^2}{v^2}\, dA = \frac{1}{4\ell_\mathrm{P}^2}\int_\Sigma f(r)\, dA\,. \tag{28}$$

Far from the black hole, $f \to 1$ and $\mathcal{I}$ reduces to the area in Planck units, consistent with the holographic bound. At the horizon, $f \to 0$ and $\mathcal{I} \to 0$: the horizon carries zero causal information. In the interior, $\langle|\varphi|\rangle = 0$ and $\mathcal{I} = 0$ identically. The interior has no causal information content.

8.2 Information nullification at the horizon

When matter crosses the horizon, it passes from a region of nonzero causal information ($\mathcal{I} > 0$) to a region of zero causal information ($\mathcal{I} = 0$). Its information is not transferred to the interior; it is *nullified*—converted from causal content to nothingness.

The Bekenstein–Hawking entropy [10]

$$S = \frac{k_\mathrm{B}\, A}{4\ell_\mathrm{P}^2} \tag{29}$$

counts the number of Planck-area cells on the horizon. In our framework, each cell represents one unit of causal information that has been nullified: one Planck area's worth of causal condensate that has been destroyed. *Entropy is the tally of nullified causality.*

8.3 The information paradox

The black hole information paradox [8] asks: where does the information go when matter falls into a black hole? Our answer:

It goes to nothingness.

This is not evasion. It is a precise statement within our framework. The causal condensate in the exterior, which encodes and supports all causal information, is systematically destroyed at the horizon. The information does not "go" anywhere—the medium that supported it ceases to exist. Asking where the information went is like asking where the shape of an ice cube went after it melted: the medium that defined it (solid ice / causal condensate) has undergone a phase transition, and the property (shape / information) has no meaning in the new phase (liquid water / nothingness).

The Hawking radiation [7] is the thermal signature of this phase transition. It carries energy away from the horizon but does not carry coherent information, just as the thermal radiation from a melting ice cube carries energy but not the ice's shape. This is consistent with Hawking's original conclusion that the radiation is thermal and does not encode the information of the infalling matter.

9 The Causality Destruction Theorem

9.1 Total exterior causal content

Definition 9.1 (Total exterior causal content)**.** *The total causal content of the exterior on a spacelike slice is*

$$\mathcal{C}(M) = 4\pi v^2 \int_{r_\mathrm{s}}^{R} \frac{r^2 f(r)}{\sqrt{f(r)}}\, dr = 4\pi v^2 \int_{r_\mathrm{s}}^{R} r^2 \sqrt{f(r)}\, dr\,, \tag{30}$$

where R is an infrared cutoff (e.g., the cosmological horizon) and $r_\mathrm{s} = 2GM/c^2$.

9.2 Monotonic decrease

Theorem 9.2 (Causality destruction)**.** *If the black hole mass increases ($\dot{M} > 0$), the total exterior causal content decreases:*

$$\boxed{\dot{M} > 0 \implies \frac{d\mathcal{C}}{dt} < 0\,.} \tag{31}$$

Proof. We have $\mathcal{C} = \mathcal{C}(r_\mathrm{s})$ where $r_\mathrm{s} = 2GM/c^2$. Computing $d\mathcal{C}/dr_\mathrm{s}$:

$$\frac{d\mathcal{C}}{dr_\mathrm{s}} = -4\pi v^2\, r_\mathrm{s}^2 \underbrace{\sqrt{f(r_\mathrm{s})}}_{=0} + 4\pi v^2 \int_{r_\mathrm{s}}^{R} r^2 \frac{\partial}{\partial r_\mathrm{s}} \sqrt{f(r)}\, dr\,. \tag{32}$$

The boundary term vanishes since $f(r_\mathrm{s}) = 0$. The integral term is

$$\frac{\partial}{\partial r_\mathrm{s}} \sqrt{f(r)} = \frac{\partial}{\partial r_\mathrm{s}} \sqrt{1 - r_\mathrm{s}/r} = \frac{-1}{2r\sqrt{f(r)}} < 0\,, \tag{33}$$

for all $r > r_\mathrm{s}$. Therefore

$$\frac{d\mathcal{C}}{dr_\mathrm{s}} = -2\pi v^2 \int_{r_\mathrm{s}}^{R} \frac{r}{\sqrt{f(r)}}\, dr < 0\,. \tag{34}$$

Since $\dot{r_s} = (2G/c^2)\dot{M} > 0$ when $\dot{M} > 0$, we have

$$\frac{d\mathcal{C}}{dt} = \frac{d\mathcal{C}}{dr_s} \cdot \dot{r_s} < 0\,. \tag{35}$$

□

9.3 Interpretation

As the horizon grows, the causal condensate is weakened throughout the *entire exterior*—not only near the horizon. The factor $\sqrt{f(r)} = \sqrt{1 - r_s/r}$ decreases at every fixed r when r_s increases. Every point in the exterior loses some causal content. The black hole is not merely eating spacetime at its boundary; it is diluting the causal fabric of the entire exterior.

The second law of black hole mechanics ($\delta A \geq 0$, Ref. [9]), combined with our framework, states:

> *The total amount of nothingness in the universe never decreases. Every increase in horizon area represents an irreversible conversion of causal spacetime into causally disconnected spacetime.*

10 Duality with Hawking Radiation

The Hawking mass-loss rate for a Schwarzschild black hole [7] is

$$\dot{M}_{\text{Hawk}} = -\frac{\hbar\, c^6}{15360\pi\, G^2\, M^2}\,. \tag{36}$$

The causal mass creation rate (25) is

$$\dot{M}_{\text{create}} = +\frac{\beta\, c^8}{16\, G^2\, M^2}\,. \tag{37}$$

Both scale as M^{-2} with opposite signs. The net evolution is

$$\dot{M}_{\text{net}} = \frac{c^6}{G^2 M^2}\left(\frac{\beta\, c^2}{16} - \frac{\hbar}{15360\pi}\right). \tag{38}$$

Mass creation dominates when $\beta > \beta_{\text{crit}} \equiv \hbar/(960\pi c^2)$.

This parallel scaling is not a coincidence. Both processes originate at the horizon and are governed by the surface gravity:

- **Mass creation:** The causal condensate is destroyed at the horizon. The gradient energy of the collapsing condensate is captured as gravitational mass.
- **Hawking radiation:** The same condensate collapse, viewed quantum mechanically, produces particle–antiparticle pairs at the boundary. The outgoing particle carries energy away.

The two rates are related by

$$\frac{\dot{M}_{\text{create}}}{|\dot{M}_{\text{Hawk}}|} = \frac{15360\pi\,\beta\,c^2}{\hbar}\,, \tag{39}$$

a dimensionless ratio determined by β relative to $\hbar/c^2$.

If $\beta \geq \beta_{\text{crit}}$, the black hole grows monotonically. In this regime, Hawking radiation is the *thermal byproduct* of the causal Higgs mechanism—the waste heat of the nothingness factory—while the dominant physical process is the irreversible conversion of causal spacetime into nothingness with accompanying mass generation.

11 Summary of Results

Result	**Equation**
Spatial causal disconnectedness of interior	Prop. 2.1
Causal coherence time	$\tau_{\text{coh}}(r) \to 0$ as $r \to 0$
Dissolution radius for object of size L	$c\,\tau_{\text{coh}}(r_{\text{diss}}) = L$
Causal VEV profile	$\langle\lvert\varphi\rvert\rangle = v\sqrt{f(r)}\,\Theta(r - r_{\text{s}})$
Gauge boson mass	$m_{\mathcal{A}}(r) = g_{\mathcal{C}} v\sqrt{2f(r)}\,\Theta(r - r_{\text{s}})$
Mass creation rate	$\dot{M} = \beta c^8/(16G^2M^2) \propto M^{-2}$
Horizon growth law	$r_{\text{s}}(t) \propto t^{1/3}$ (late time)
Causality destruction	$d\mathcal{C}/dt < 0$ whenever $\dot{M} > 0$
Critical coupling	$\beta_{\text{crit}} = \hbar/(960\pi c^2)$

12 Discussion

12.1 What is established vs. what is proposed

We distinguish clearly between established results and proposals.

Established (following from the Schwarzschild metric alone):

- Proposition 2.1: spatial causal disconnectedness of the interior.
- The causal coherence time $\tau_{\text{coh}}(r)$ and its vanishing at the singularity.
- The dissolution condition (5) for extended objects.
- Theorem 9.2: monotonic decrease of exterior causal content (given any mechanism that increases r_{s}).

Proposed (constituting our theoretical framework):

- The identification of nothingness with causal disconnectedness (theorem 4.1).
- The $U(1)_{\mathcal{C}}$ gauge theory and the causal Higgs mechanism.
- The geometric coupling postulate (postulate 2).
- The mass creation law (24) and the coupling constant β.
- The reinterpretation of entropy as nullified causality.

12.2 Observational consequences

(i) **Primordial black holes.** If $\beta > \beta_{\text{crit}}$, small primordial black holes grow rather than evaporate. This modifies the standard constraints on their mass spectrum and abundance, and their potential role as a dark matter component.

(ii) **Supermassive black holes.** The $\dot{M} \propto M^{-2}$ scaling predicts that supermassive black holes have negligible mass creation rates, consistent with their observed quiescence.

(iii) **Gravitational wave signatures.** During a merger, the sudden increase in horizon area represents a discrete jump in causality destruction. The quasi-normal mode spectrum of the remnant may encode the dynamics of the causal condensate collapse.

12.3 Open problems

(i) The value of β is not determined by our framework. A microscopic derivation from quantum gravity would fix it.

(ii) The extension to Kerr and Reissner–Nordström spacetimes, where the causal structure near inner and outer horizons differs qualitatively, is nontrivial.

(iii) The causal gauge field $\mathcal{A}_\mu$ may have a geometric interpretation—for instance, as a component of the connection in a Kaluza–Klein or teleparallel framework. This remains to be explored.

(iv) The cosmological implications—the total rate of causality destruction across all black holes in the observable universe, and its relation to the cosmological constant—are unexplored.

13 Conclusion

The Schwarzschild interior is a region where equal-time spatial hypersurfaces are Riemannian: no two distinct points on them can be connected by a causal curve. This is a fact of Schwarzschild geometry. Its consequence is that spatially extended objects—everything from galaxies to electrons—cannot maintain the causal coherence required for their existence.

We have identified this causal disconnectedness with nothingness: not the absence of spacetime, but the absence of the causal infrastructure that allows physical objects to be. We have proposed that the transition from causal spacetime (exterior) to nothingness (interior) is governed by a Higgs mechanism in which a causal scalar field loses its condensate at the horizon, destroying the causal mediator's ability to propagate causal influence.

The central physical picture is this: the event horizon is a phase boundary. On one side, the causal condensate is nonzero, the causal mediator is effective, objects can exist, and physics happens. On the other side, the condensate is zero, the mediator is inert, objects dissolve, and nothing happens. Mass is generated at this boundary because the destruction of the condensate costs energy, and this energy is gravitational mass.

Nature, through the event horizon, converts being into nothingness. The second law of black hole mechanics ensures that this conversion is irreversible. The universe, one Planck area at a time, is being returned to causal silence.

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